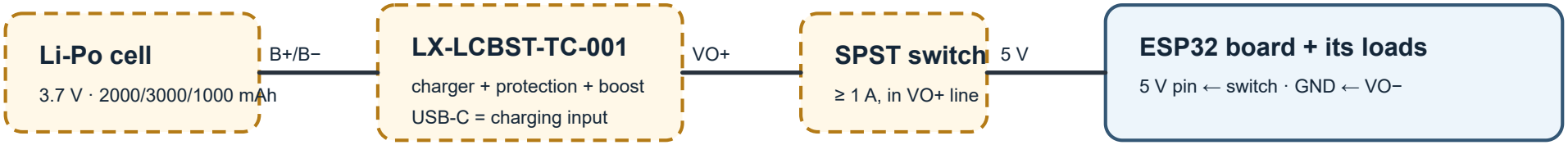
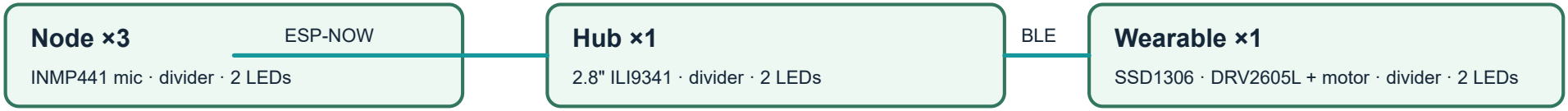
CryNet battery-powered system — wiring overview

Same power chain in every unit; signal wiring differs per unit (pages 2–4)



Signals, per unit (see following pages):



Rules every unit follows:

1 · Switch OFF before charging — the battery still charges on B+/B−.

2 · Set the boost trimpot to 5.1 V with a meter BEFORE connecting any ESP32.

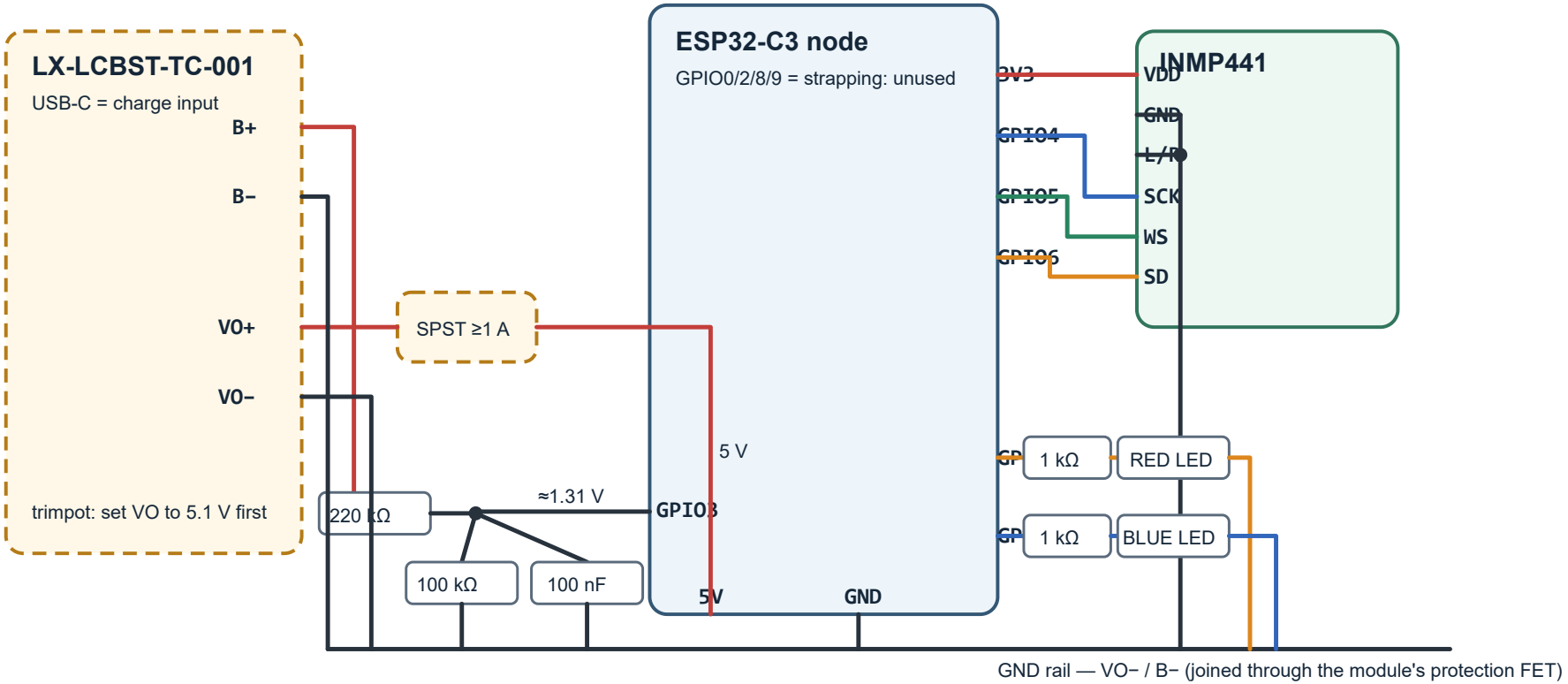
3 · Never power a board from USB and the boost output at the same time.

Dashed = proposed, bench-validation pending · Solid green = verified on the live system.

No signal wire exists between nodes and hub — the link is wireless. Each unit is electrically independent; never join their batteries or grounds. Read with SYSTEM_WIRING.md §5–§6.

2 / Cot node — complete wiring (one diagram per node, ×3)

ESP32-C3 (SuperMini now, Pro Mini planned) + INMP441 + LX-LCBST + switch + divider + 2 LEDs



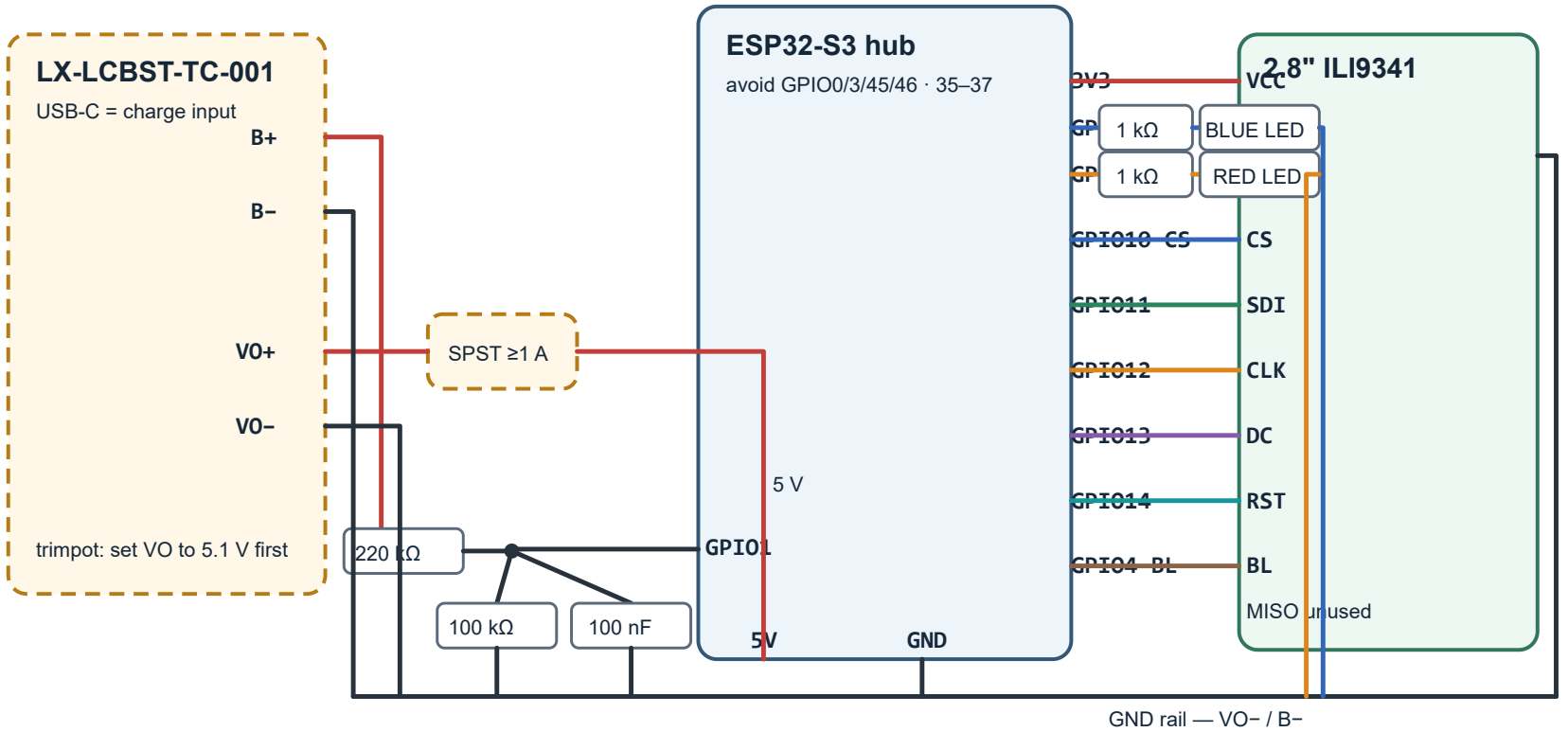
Verified on the live system: mic wiring (3V3/GND/L/R→GND, GPIO4 SCK, GPIO5 WS, GPIO6 SD) and the node firmware. GPIO1 stays spare.

Proposed, needs bench validation: the whole battery chain. GPIO3 ADC divider ≈13 µA drain. BLUE LED = link (lights only on a fresh heartbeat ACK from the hub, slow-blink otherwise). RED LED = SOC (off / slow-blink low / fast-blink critical). Firmware for divider + LEDs + ACK does not exist yet — branch work.

Wires may cross without joining (dots mark real junctions). Pin names, not header positions. Repeat identically for nodes 2 and 3.

3 / Main hub — complete wiring (×1)

ESP32-S3-DevKitC-1 N16R8 + 2.8" ILI9341 + LX-LCBST + switch + divider + 2 LEDs



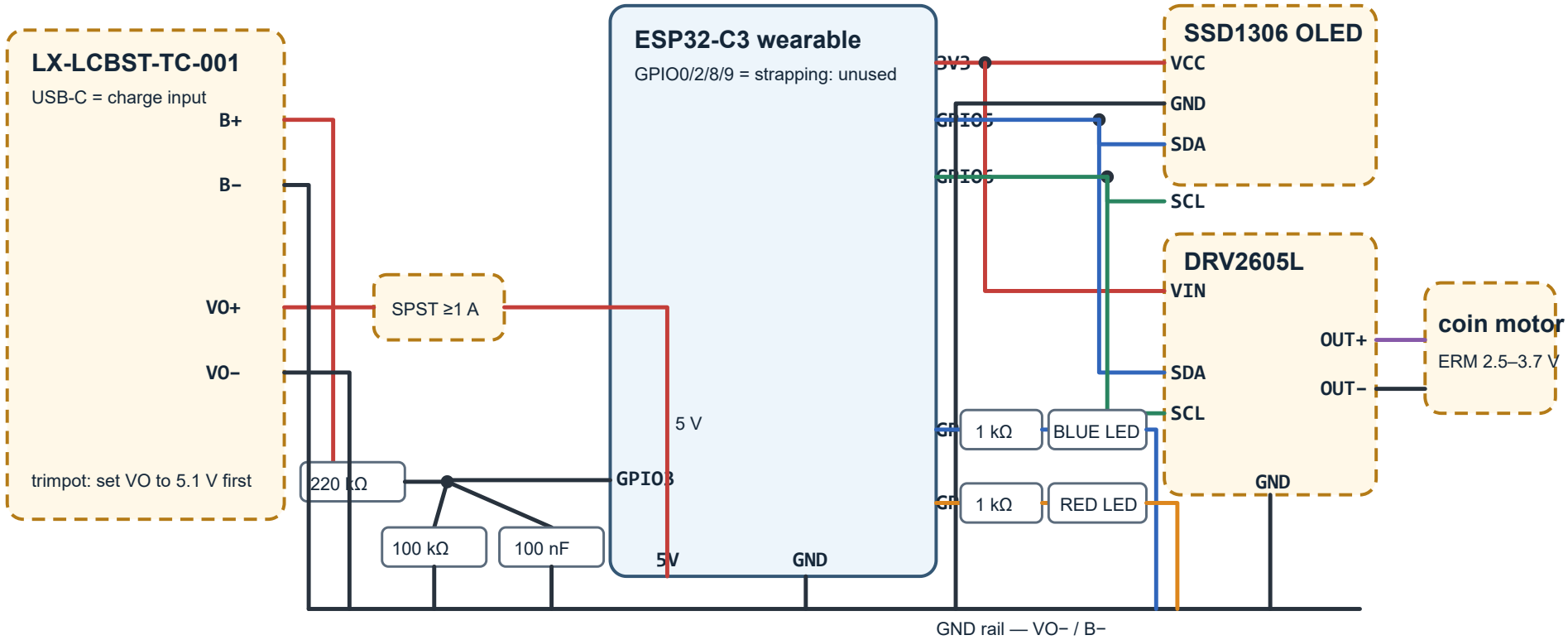
Verified on the live system: TFT signals (CS 10, MOSI 11, SCK 12, DC 13, RST 14, BL 4, VCC 3V3, GND) and the hub firmware.

Proposed: battery chain, divider on GPIO1 (ADC1_CH0 — works while Wi-Fi/BLE run), BLUE LED on GPIO2 (BLE client connected), RED LED on GPIO17 (SOC). The hub's peak current with TFT + radios is unmeasured — confirm the LX-LCBST holds 5.1 V during bursts before trusting it here.

GPIO35–37 are the Octal PSRAM bus — never wire them. The TFT wiring is the recorded working one for the existing module.

4 / Wearable — complete wiring (×1)

ESP32-C3 SuperMini (reassigned from the node) + SSD1306 + DRV2605L + motor + LX-LCBST + divider + 2 LEDs

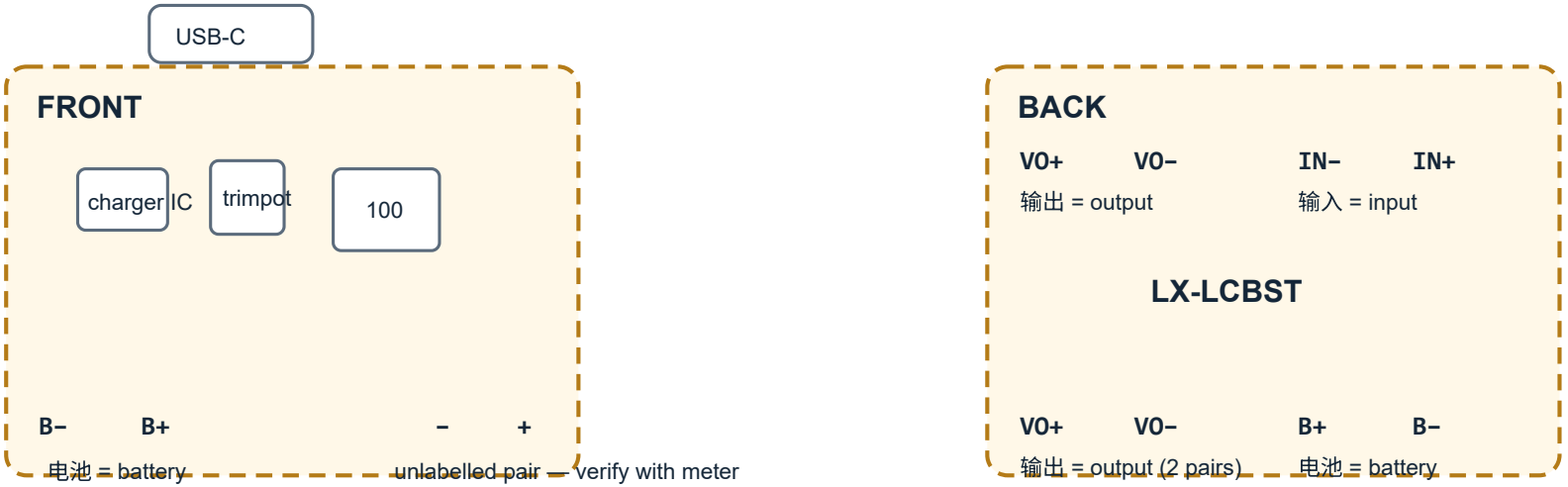


Proposed, nothing built: OLED + DRV2605L power from the SuperMini's 3V3 pin (never 5 V — the DRV breakout's pull-ups would lift I²C to 5 V). Check the board's regulator budget: radio peaks + OLED + motor share it. Motor across OUT+/OUT− only — no MOSFET, no flyback diode. Divider on GPIO3; BLUE LED GPIO1 (BLE link), RED LED GPIO4 (SOC).

Wearable firmware (BLE client, OLED, haptics) does not exist yet. Node role moves to the Pro Mini boards; this SuperMini becomes the wearable.

5 / LX-LCBST-TC-001 — pad map and bench procedure

From the user's front-and-back photograph · Chinese silk text translated



Bench procedure, before any ESP32 is connected:

1 · Battery on B+/B− → trimpot until VO+/VO− reads 5.1 V (meter, no load). 2 · USB-C in: measure charge current (series meter) — the cell must allow it.

3 · Confirm charge ends at 4.2 V. 4 · Switch off: confirm the load is dead and measure battery drain (boost quiescent is unknown). 5 · Run one node; compare with the USB baseline.

No T pad on this board — the listing's T-shorting notes belong to other variants. Never short anything.

Unknown until measured: actual charge current (program resistor unreadable in the photo), boost quiescent drain when off, protection thresholds, and what the front − / + pair connects to. Buy ONE module for validation; only then buy the remaining three (plus the wearable's, pending the same check).